College of Computing

Department of Information Technology

Event-Driven Programming
Group Project

Title:School Fee Managemet System

Course Code: ITec3052

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Application Information to login on application

For Admin

Username:Admin

Passward: admin12345

For Staff

Username:Staff

Passward: staff1234

You can also view staff roles functionalities by creating self-accounts through the admin.

Introduction

The School Fee Management System is a desktop and web-based application developed to automate school fee management activities. The system helps schools manage student information, process payments, generate receipts, and prepare financial reports efficiently. The project reduces manual paperwork, improves data accuracy, and increases security in financial management.

The system was developed using C#, ASP.NET Core Web API, Windows Forms, Entity Framework Core, and SQL Server Database. These technologies work together to create a reliable and user-friendly application.

Core Concepts of the Project Both Admin and Staff page

Authentication

Authentication is the process of verifying users before accessing the system. Users must enter valid usernames and passwords to log in. This helps protect sensitive school financial information from unauthorized access.

Authorization

Authorization controls user permissions after login. The system uses role-based access control with two main roles:

- Admin
- Staff

Admins have full access to manage the system, while staff members have limited access such as recording payments and printing receipts.

Student Management

The student management module stores and manages student information including:

- Student name
- Class
- Gender
- Contact details

The system allows users to add, update, delete, and search student records.

Fee Category Management

Fee categories organize different types of school fees such as:

- Tuition Fee
- Registration Fee
- Library Fee
- Laboratory Fee

This helps schools manage different payment types properly.

Fee Structure Management

Fee structure defines the amount students must pay depending on their class or fee category. It helps standardize payment amounts within the school.

Payment Processing

The payment module records student fee payments and stores transaction details securely in the database. The system automatically generates receipt numbers and reference numbers during payment processing.

Receipt Generation

After successful payment, the system automatically generates receipts containing:

- Student information
- Payment amount
- Receipt number
- Payment date
- Reference number

Receipts serve as proof of payment and improve financial accountability.

Report Generation

The report module generates financial summaries such as:

- Daily payment reports
- Monthly reports
- Student payment history
- Outstanding balance reports

Reports help administrators monitor financial activities effectively.

For Only Admin Has:

Settings Management

The Settings module allows the administrator to configure and manage system settings. This module helps customize how the system works and controls important application features.

The settings module may include:

- School information settings

- System configuration
- Password management
- User account settings
- Theme or interface settings
- Backup settings

Only the Admin can access and modify system settings. This improves system security and prevents unauthorized changes.The settings module helps maintain system organization, flexibility, and proper configuration.

Staff Management

The Staff Management module allows the Admin to manage staff accounts within the system.

The module includes:

- Add staff
- Update staff information
- Delete staff accounts
- Assign roles
- Manage usernames and passwords

Staff members use their accounts to log into the system and perform authorized activities such as recording payments and printing receipts.

This module improves user management and controls staff access properly.

Database Management

The project uses SQL Server Database to store all system information securely. The database manages students, users, payments, fee structures, and reports.Database management improves data organization, storage, and retrieval.

Entity Framework Core

Entity Framework Core is used as the Object Relational Mapper (ORM). It simplifies communication between the application and the database by allowing developers to work with C# objects instead of writing complex SQL queries.

ASP.NET Core Web API

The API acts as the communication bridge between the frontend application and the database. It processes requests, performs business logic, and returns responses to the user interface.

Windows Forms

Windows Forms provides the graphical user interface (GUI) of the application. Users interact with forms such as:

- Login Form
- Student Form
- Payment Form
- Report Form

The interface is designed to be simple and user-friendly.

CRUD Operations

CRUD operations are basic database activities used throughout the project.

- Create → Add records
- Read → Retrieve records
- Update → Modify records
- Delete → Remove records

These operations help manage all system data.

Validation

Validation checks user input before saving information to the database. It prevents invalid or incomplete data entry.

Examples include:

- Empty field validation
- Numeric validation
- Password validation

Validation improves data accuracy and reliability.

Error Handling

Error handling manages unexpected problems in the system and prevents application crashes. It helps display meaningful messages to users and improves system stability.

DataGridView

DataGridView is used to display records in table format such as student lists, payment records, and reports. It improves data visibility and organization.

Data Binding

Data binding automatically connects database data to user interface controls. It simplifies displaying information in forms and tables.

DTO (Data Transfer Object)

DTOs are used to transfer data between the frontend, API, and database securely and efficiently. They reduce unnecessary data transfer and improve performance.

Dependency Injection

Dependency Injection improves code organization by automatically providing required services and dependencies to system components.

Asynchronous Programming

Asynchronous programming allows the system to perform operations without freezing the user interface. This improves performance and responsiveness.

Layered Architecture

The system uses a three-layer architecture:

- Presentation Layer
- Business Logic Layer

- Data Access Layer

This architecture improves maintainability, scalability, and organization.

Security

The system includes security features such as authentication, authorization, validation, and protected database access to secure sensitive information.

Advantages of the Project

The School Fee Management System provides many advantages including:

- Reduces paperwork
- Saves time
- Improves accuracy
- Enhances security
- Simplifies fee management
- Generates reports quickly
- Improves financial transparency

Challenges Faced During Development

During development, several challenges were encountered such as:

- Database relationship configuration
- API connection issues
- Validation problems
- DataGridView display issues
- Authentication errors

These challenges were solved through debugging and testing.

Conclusion

The School Fee Management System successfully automates school fee management activities. The project improves efficiency, accuracy, and security by replacing manual operations with computerized processes.

Using technologies such as C#, ASP.NET Core Web API, Windows Forms, Entity Framework Core, and SQL Server, the system provides a reliable and user-friendly solution for managing student fees, payments, receipts, and financial reports effectively.